

Akshat Dikshit

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EDUCATION

SRMIST Kattankulathur

Chennai, India

B.Tech in Computer Science | CGPA: 8.4

Aug. 2022 – May 2026

- Relevant coursework: Data Structures & Algorithms, Machine Learning, Computer Vision, Database Management Systems (DBMS), Operating Systems

EXPERIENCE

Machine Learning Intern

Apr. 2026 – Present

Jio Platforms Ltd. (Reliance Industries)

Gurugram, India

- Engineered **FaceTools**: an asynchronous multi-container microservices suite of AI inference based on docker containers running decoupled deep learning models with ONNX runtime with YOLOv8 tracking and arcFace modules, utilizing parallel async workers for maximum throughput.
- Built the **FaceVision** data analysis engine to run DBSCAN clustering models across 512-dimensional vector embeddings; developed relational table formats within PostgreSQL enabling automatic grouping and high dimensional vector similarity search queries over media datasets.
- Decentralized backend analytics microservices accessed through a nginx reverse proxy gateway and decoupled react frontend interface enabled streamlined production deployments by automating GitHub actions CI/CD workflows and multi-service docker compose orchestration layers.

PROJECTS

Landslide Susceptibility Mapping | Python, Scikit-learn, GEE

github.com/Akd2510/uk_ml_major

- Developed an end-to-end predictive geospatial machine learning pipeline combining regional global landslide catalog coordinates with Google Earth engine spatial terrain data arrays (SRTM DEM, TWI, NDVI) and IMD gridded netCDF precipitation records.
- Stratified 5-fold cross-validation methods were employed to address severe class imbalance data distributions, and a balanced random forest architecture was trained and optimized. The macro **F1 score achieved was .93**, and the **out-of-bag (OOB) generalization estimate was .61**.
- Feature importance metrics were evaluated using SHAP explainability matrices; an **AUC-ROC profile of 0.663** was analyzed to illustrate boundary ambiguity in probabilistic score space while demonstrating strong per-class precision and recall for rare high susceptibility hazard zones.

ProjectTran: Transit Tracker | Next.js, Supabase, Prisma, TypeScript

github.com/Akd2510/ProjectTran

- Developed a full-stack transit tracking application utilizing next.js framework components to render synchronized route paths and live spatial updates via geometric APIs from google maps.
- Designed a normalized relational database schema within PostgreSQL utilizing prisma orm, implemented targeted b-tree indexing on frequent query columns to optimize lookups and maintain transactional data integrity.
- Integrated real-time weather data API hooks to dynamically inform routing conditions and established secure, persistent user session handling managed entirely via supabase auth pipelines.

Motion: Collaborative Workspace | React, Node.js, Firebase, Clerk Auth

github.com/Akd2510/Motion

- Architected a real-time collaborative document platform that continued to synchronize the document state of multiple users simultaneously. Ensuring low-latency updates to the document state utilizing firebase realtime database socket layers.
- I implemented debounced state management handlers and optimized local UI component re-renders to allow for smooth processing of high frequency concurrent document modifications without causing stutter in the layout.
- Created an application connection layer that tracks the state of active network connections. Prevented client data synchronization loss due to local state caching during unexpected local disconnections.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, JavaScript, TypeScript, SQL

Machine Learning & Vision: Scikit-learn, OpenCV, MediaPipe, ONNX Runtime, NumPy, Pandas, Matplotlib, SHAP

Frameworks & Tools: React, Next.js, Node.js, FastAPI, Django, REST APIs, Firebase, Supabase, Prisma, Tailwind CSS

Databases & ORM: PostgreSQL, Firebase Realtime DB, Supabase

Cloud & DevOps: AWS (ML Foundations), GCP (BigQuery, Earth Engine), Docker, Nginx, GitHub Actions, CI/CD, Git

Operating Systems: Linux (Arch, Fedora), Windows

CERTIFICATIONS

Google Cloud Data Analytics (BigQuery, Looker Studio)

2024

AWS Machine Learning Foundations (SageMaker, ML Workflows)

2024

Cisco Networking Basics

2023